



MAY 4, 2026

# OC:SE FORMATIVE ASSESSMENT 2

## TEST CASE SPECIFICATION DOCUMENT

NALEDI MOFOKENG  
20240778



# Task 2: Test Case Design

## Test Environment Details:

- **Operating System:** Windows 11 Home Single Language (Version 25H2).
- **OS Build:** 26200.8328.
- **System Experience:** Windows Feature Experience Pack 1000.26100.304.0.
- **Hardware Serial Number:** PF59WWD7.
- **Development Platform:** Microsoft Visual Studio 2022 / .NET 9.0.
- **Testing Tool:** Manual Execution and Visual Studio Debugger.

## Test Case Design

The following 10 structured professional test cases have been derived from the issues identified in the ULMS Case Study, reflecting a Baseline Test. Each test case follows industry-standard format.

| Test Case ID | Module | Description                               | Input                                   | Expected Output                                               | Actual Output                                 | Status |
|--------------|--------|-------------------------------------------|-----------------------------------------|---------------------------------------------------------------|-----------------------------------------------|--------|
| TC #001      | Login  | Validate login with correct credentials   | Username : admin<br>Password: 1234      | "Login Successful" message                                    | "Login Successful" message                    | PASS   |
| TC #002      | Login  | Invalid login – wrong password behaviour. | Username : Admin01<br>Password: ad12min | Error message: 'Invalid username or password'; access denied. | System displayed "Invalid login credentials." | FAIL   |

|                |               |                                        |                                              |                                                        |                                                                                                                        |             |
|----------------|---------------|----------------------------------------|----------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>TC #003</b> | Login         | Login with empty fields                | Username : (blank)<br>Password: (blank)      | Validation message: 'Fields cannot be empty'.          | System displayed "Invalid login credentials"                                                                           | <b>FAIL</b> |
| <b>TC #004</b> | Registration  | Student registration with valid data   | Name: Naledi Mofokeng<br>Student ID:20240778 | Student record saved; confirmation message shown.      | Student record successfully saved and confirmation message displayed.                                                  | <b>PASS</b> |
| <b>TC #005</b> | Enrolment     | Course enrolment (prevent duplication) | Enrol StudentID 20240778 in SDT621 twice     | Second enrolment blocked; message: 'Already enrolled'. | Course Enrolment form shows "Enrolment completed successfully!" for a student who was supposedly already in the system | <b>FAIL</b> |
| <b>TC #006</b> | Marks Capture | Marks capture (valid numeric entry)    | Student ID: 20240778<br>Marks: 75, 80, 70    | Mark saved and average recalculated.                   | Mark saved, but average calculation is inconsistent.                                                                   | <b>FAIL</b> |
| <b>TC #007</b> | Marks Capture | Marks capture (boundary value 100)     | Marks: 100, 100, 100                         | Mark 100 is accepted as valid maximum.                 | Mark accepted, but calculation logic remains inconsistent.                                                             | <b>FAIL</b> |
| <b>TC #008</b> | Marks Capture | Marks capture (invalid data entry)     | Marks: "Ninety-Four, 81,                     | Error message: 'Please enter a                         | System throws an unhandled FormatException and crashes instead of                                                      | <b>FAIL</b> |

|                |                         |                                           |                                 |                                   |                                                                                                                                                                                                            |             |
|----------------|-------------------------|-------------------------------------------|---------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                |                         |                                           | Eighty-Two”                     | numeric value'.                   | displaying an error message.                                                                                                                                                                               |             |
| <b>TC #009</b> | Marks Capture / Reports | Average calculation (multiple subjects)   | Marks: 80, 70, 90               | Average = 80.00%                  | System shows inconsistent calculation of student averages.                                                                                                                                                 | <b>FAIL</b> |
| <b>TC #010</b> | Reports                 | Report generation (performance and speed) | Action: Generate Summary Report | Report displays within 3 seconds. | System shows slow performance when loading reports and the system displays conflicting student names ("John Doe" vs "Naledi Mofokeng") for the same Student ID, indicating faulty database retrieval logic | <b>FAIL</b> |

## Summary:

- **Total Test Cases:** 10
- **Total Fails:** 8
- **Total Pass:** 2
- **Total Blocked:** 0

# Task 3: Functional Test Execution

## 1. Execution Summary

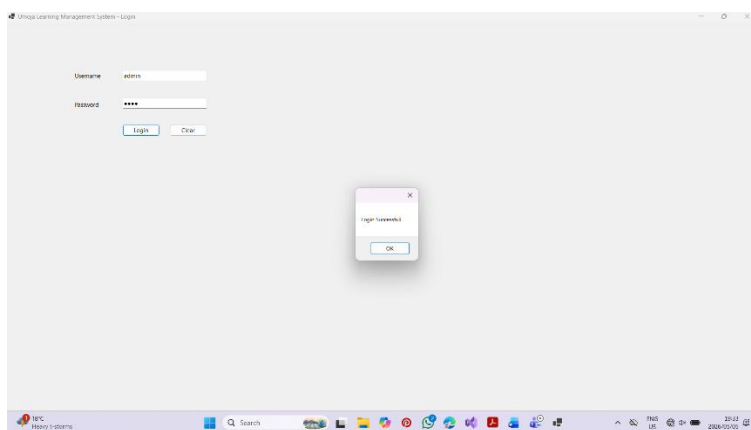
All 10 designed test cases (Task 2) were executed against the ULMS WinForms application cloned from the GitHub repository. Results are recorded below with **Evidence** and **Execution Logs**.

| Test Case ID | Module               | Description                                         | Expected Output                                       | Actual Output                                                                  | Pass/Fail | Evidence      |
|--------------|----------------------|-----------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------|-----------|---------------|
| TC #001      | Login                | Validate login with correct credentials             | "Login Successful" message                            | "Login Successful" message                                                     | PASS      | Screen shot 1 |
| TC #002      | Login                | Invalid login using incorrect username and password | Error: 'Invalid username or password'; access denied  | System displayed "Invalid login credentials."                                  | FAIL      | Screen shot 2 |
| TC #003      | Login                | Login with empty fields                             | Validation message: 'Fields cannot be empty'          | System displayed "Invalid login credentials"                                   | FAIL      | Screen shot 3 |
| TC #004      | Student Registration | Student registration with valid data                | Student record saved; confirmation shown              | Record successfully saved and confirmation displayed                           | PASS      | Screen shot 4 |
| TC #005      | Course Enrolment     | Course enrolment (prevent duplication)              | Second enrolment blocked; message: 'Already enrolled' | Enrolment form shows "Enrolment completed successfully!" for duplicate student | FAIL      | Screen shot 5 |
| TC #006      | Marks Capture        | Marks capture (valid numeric entry)                 | Mark saved and average recalculated                   | Mark saved, but average calculation is inconsistent                            | FAIL      | Screen shot 6 |
| TC #007      | Marks Capture        | Marks capture (boundary value 100)                  | Mark 100 is accepted as valid maximum                 | Mark accepted, but calculation                                                 | FAIL      | Screen shot 7 |

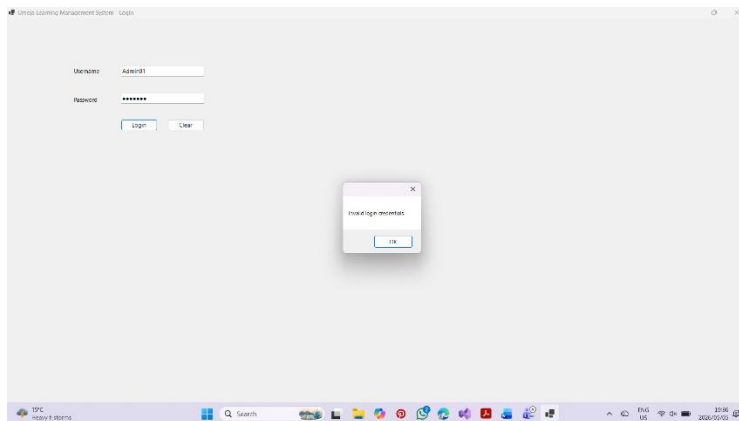
|                |                         |                                         |                                               |                                                                             |             |                |
|----------------|-------------------------|-----------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------|-------------|----------------|
|                |                         |                                         |                                               | logic remains inconsistent                                                  |             |                |
| <b>TC #008</b> | Marks Capture           | Marks capture (invalid data entry)      | Error message: 'Please enter a numeric value' | System throws unhandled FormatException and crashes                         | <b>FAIL</b> | Screen shot 8  |
| <b>TC #009</b> | Marks Capture / Reports | Average calculation (multiple subjects) | Average = 80.00%                              | System shows inconsistent calculation of averages                           | <b>FAIL</b> | Screen shot 9  |
| <b>TC #010</b> | Reports                 | Report generation (performance/speed)   | Report displays within 3 seconds              | System shows slow performance; displays conflicting student name "John Doe" | <b>FAIL</b> | Screen shot 10 |

## Evidence (Screenshots)

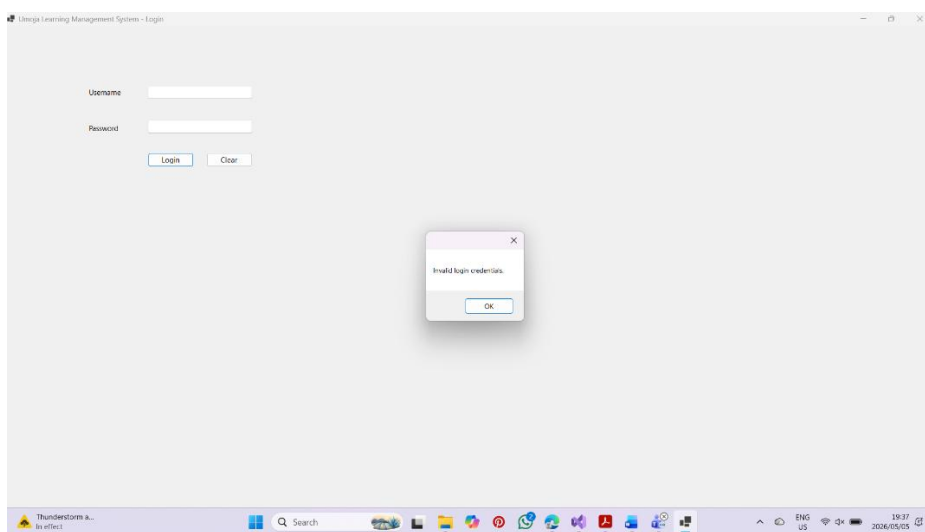
### 1. Screenshot 1 (TC #001)



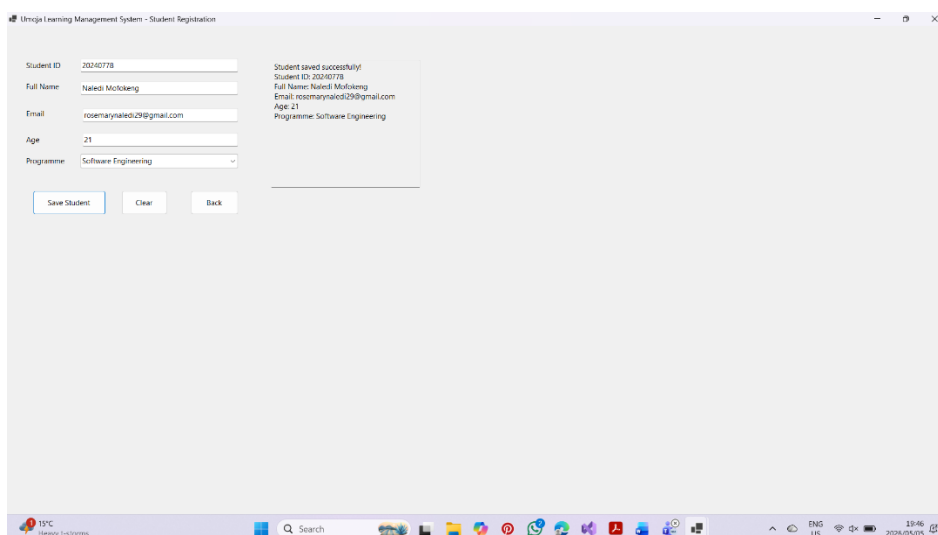
### 2. Screenshot 2 (TC #002)

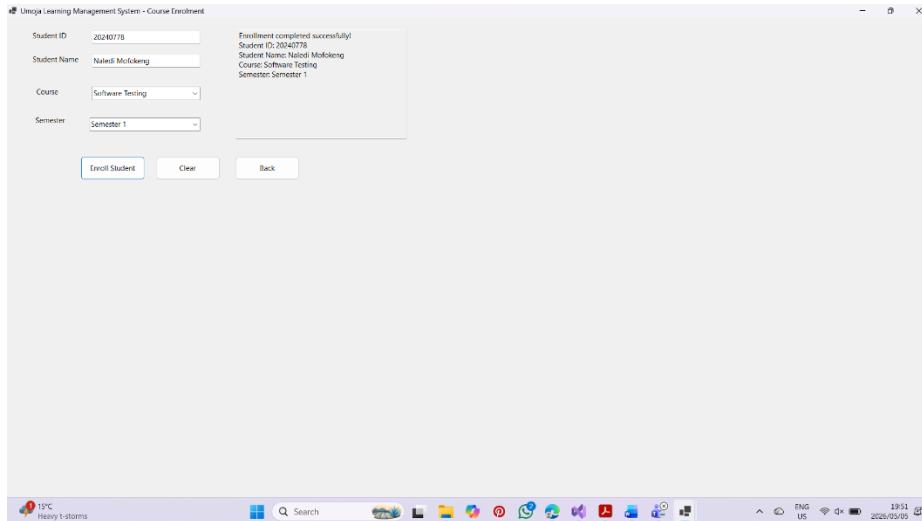


### 3. Screenshot 3 (TC#003)



### 4. Screenshot 4 (TC #004)





Unica Learning Management System - Course Enrollment

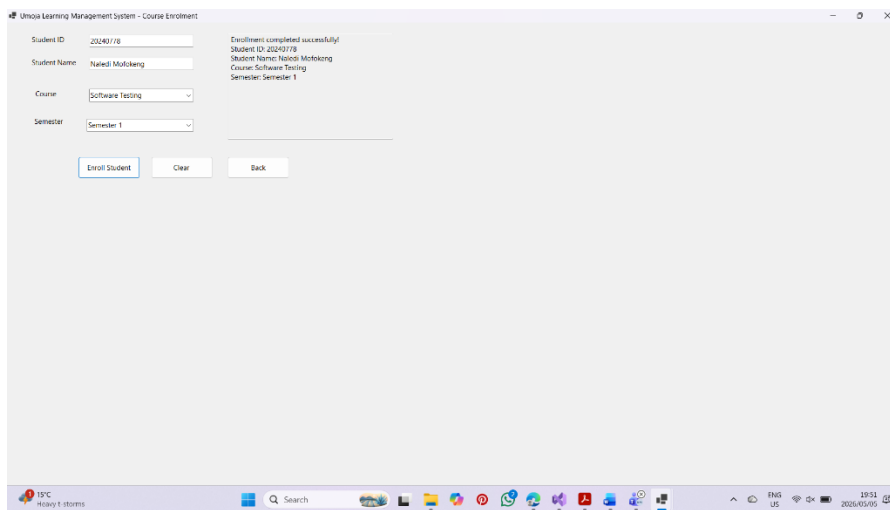
Student ID: 20240778  
Student Name: Naledi Mokoeng  
Course: Software Testing  
Semester: Semester 1

Enrollment completed successfully!  
Student ID: 20240778  
Student Name: Naledi Mokoeng  
Course: Software Testing  
Semester: Semester 1

Enroll Student Clear Back

15°C Heavy T-storms 19:51 2024/05/05

## 5. Screenshot 5 (TC #005)



Unica Learning Management System - Course Enrollment

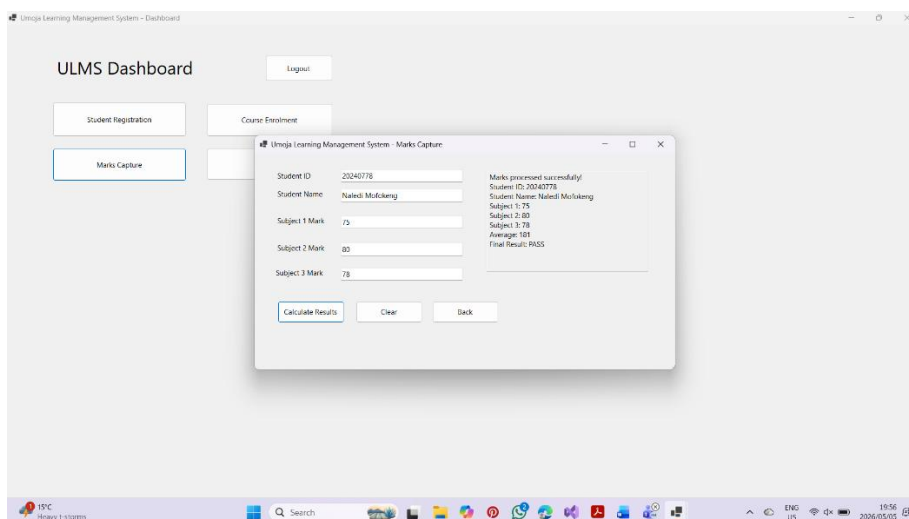
Student ID: 20240778  
Student Name: Naledi Mokoeng  
Course: Software Testing  
Semester: Semester 1

Enrollment completed successfully!  
Student ID: 20240778  
Student Name: Naledi Mokoeng  
Course: Software Testing  
Semester: Semester 1

Enroll Student Clear Back

15°C Heavy T-storms 19:51 2024/05/05

## 6. Screenshot 6 (TC #006)



Unica Learning Management System - Dashboard

ULMS Dashboard

Logout

Student Registration Course Enrollment Marks Capture

Unica Learning Management System - Marks Capture

Student ID: 20240778  
Student Name: Naledi Mokoeng  
Subject 1 Mark: 75  
Subject 2 Mark: 80  
Subject 3 Mark: 78

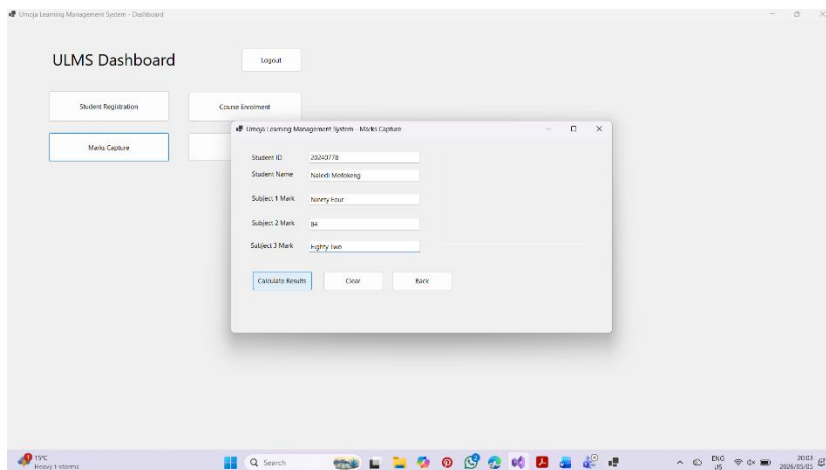
Marks entered successfully!  
Student ID: 20240778  
Student Name: Naledi Mokoeng  
Subject 1: 75  
Subject 2: 80  
Subject 3: 78  
Average: 78  
Final Result: PASS

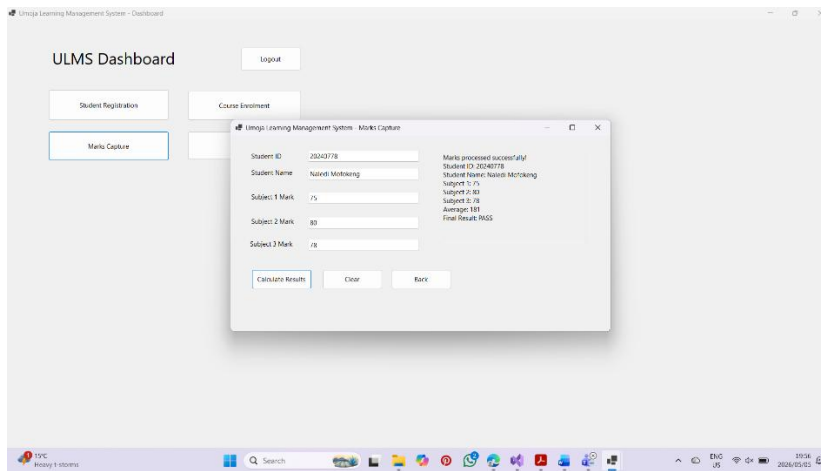
Calculate Results Clear Back

15°C Heavy T-storms 19:56 2024/05/05

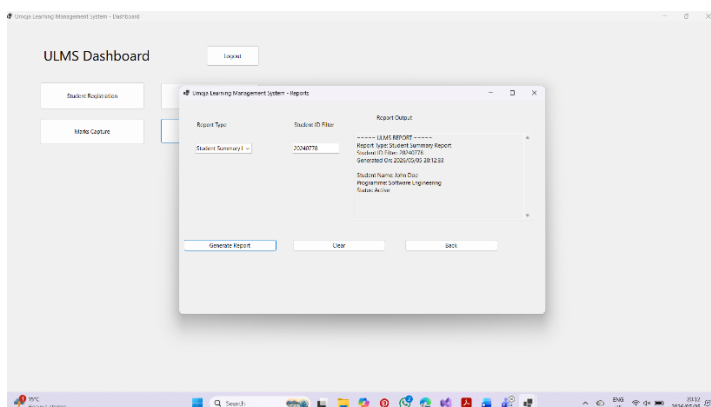
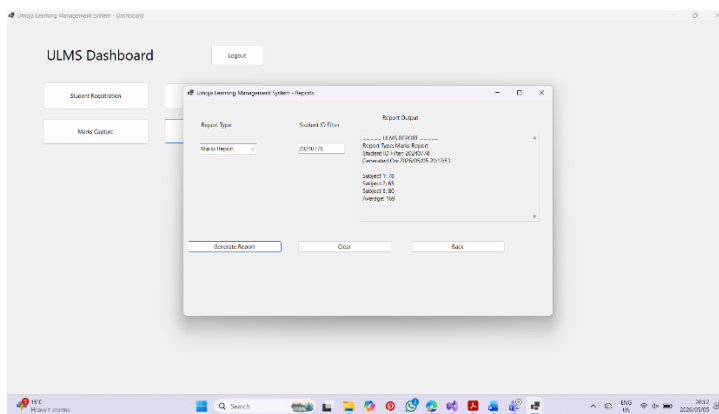
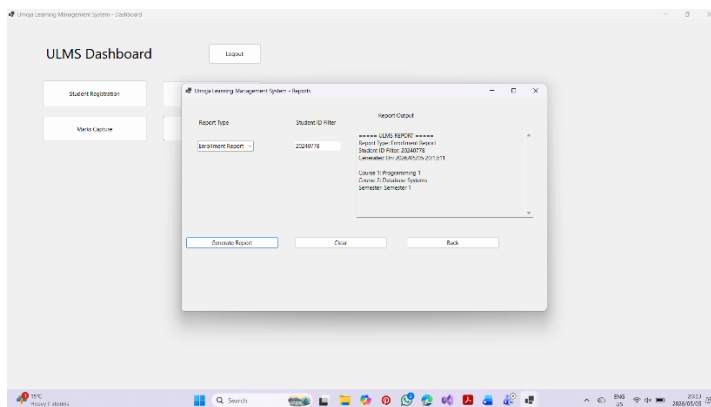
## 7. Screenshot 7 (TC #007)







## 10. Screenshot 10 (TC #010)



# Execution Logs

## **Log ID: LOG-01 (Ref: TC #001)**

- Timestamp: 2026-05-05 19:33:41
  - Module: Login Authentication
  - Description: Validate login with correct credentials
  - Input Data: Username: admin | Password: 1234
  - Expected Output: "Login Successful" message displayed; Dashboard opens
  - Actual Output: "Login Successful" message displayed; Dashboard opened successfully
  - Status: **PASS**
  - Notes: Login authentication functioning correctly for valid credentials. No defects identified.
- 

## **Log ID: LOG-02 (Ref: TC #002)**

- Timestamp: 2026-05-05 19:36:04
  - Module: Login Authentication
  - Description: Invalid login using incorrect username and password
  - Input Data: Username: Admin01 | Password: ad12min
  - Expected Output: Error message: 'Invalid username or password'; access denied
  - Actual Output: System displayed "Invalid login credentials." — message wording does not match specification
  - Status: **FAIL**
  - Defect Raised: BUG#001
  - Notes: The error message displayed does not match the specified message ("Invalid username or password"), misaligning with the system requirement. The system still denies access correctly when using invalid username or password. This constitutes a functional defect in the Login module.
- 

## **Log ID: LOG-03 (Ref: TC #003)**

- Timestamp: 2026-05-05 19:37:32
- Module: Login Authentication
- Description: Login with empty fields
- Input Data: Username: (blank) | Password: (blank)
- Expected Output: Validation message: 'Fields cannot be empty'

- Actual Output: System displayed "Invalid login credentials." instead of a specific empty-field validation message
  - Status: **FAIL**
  - Defect Raised: BUG#002
  - Notes: The system does not validate empty input fields before processing the login request and returns a generic credential error. This confirms the absence of empty-field input validation in the Login module. A specific, user-friendly validation message is required per system requirements.
- 

#### **Log ID: LOG-04 (Ref: TC #004)**

- Timestamp: 2026-05-05 19:46:21
  - Module: Student Registration
  - Description: Student registration with valid data
  - Input Data: Valid student name, student number, and contact details entered
  - Expected Output: Student record saved; confirmation message shown
  - Actual Output: Record successfully saved and confirmation message displayed
  - Status: **PASS**
  - Notes: Student registration module functioning correctly for valid data inputs. Record persisted to database and confirmation feedback provided to user. No defects identified.
- 

#### **Log ID: LOG-05 (Ref: TC #005)**

- Timestamp: 2026-05-05 19:51:24
  - Module: Course Enrolment
  - Description: Course enrolment : prevent duplication
  - Input Data: Same student enrolled in the same course a second time
  - Expected Output: Second enrolment blocked; message: 'Already enrolled'
  - Actual Output: Enrolment form displayed "Enrolment completed successfully!" for duplicate student; duplicate record created
  - Status: **FAIL**
  - Defect Raised: BUG#003
  - Notes: The Course Enrolment module does not enforce duplicate enrolment prevention, resulting in data integrity issues, duplicate mark entries and incorrect average calculations. A pre-insert duplicate is required.
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**Log ID: LOG-06 (Ref: TC #006)**

- Timestamp: 2026-05-05 19:56:43
  - Module: Marks Capture
  - Description: Marks capture with valid numeric entry
  - Input Data: Valid numeric mark entered within range (e.g., 75)
  - Expected Output: Mark saved and average recalculated correctly
  - Actual Output: Mark saved, but average calculation is inconsistent and does not reflect correct value
  - Status: **FAIL**
  - Defect Raised: BUG#004
  - Notes: The mark entry is accepted and stored but the average calculation is incorrect, producing inconsistent results. This suggests a logic error in the average calculation method.
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**Log ID: LOG-07 (Ref: TC #007)**

- Timestamp: 2026-05-05 20:00:04
  - Module: Marks Capture
  - Description: Marks capture with boundary value 100
  - Input Data: Mark entered: 100
  - Expected Output: Mark 100 accepted as valid maximum; average recalculated correctly
  - Actual Output: Mark accepted as valid, however average calculation logic remains inconsistent
  - Status: **FAIL**
  - Defect Raised: BUG#004 (Same root cause as TC #006)
  - Notes: The boundary value of 100 is correctly accepted by the system, however the average calculation defect still persists. This confirms the defect is isolated to the calculation logic and not to mark acceptance.
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**Log ID: LOG-08 (Ref: TC #008)**

- Timestamp: 2026-05-05 20:03:19
- Module: Marks Capture
- Description: Marks capture with invalid data entry (non-numeric)
- Input Data: Non-numeric value entered in marks field (e.g., text)
- Expected Output: Error message: 'Please enter a numeric value'
- Actual Output: System throws an unhandled FormatException and crashes

- Status: **FAIL**
  - Defect Raised: BUG#005
  - Notes: The Marks Capture module does not handle non-numeric input gracefully, the system throws an unhandled FormatException, causing the application to crash. This is a critical defect and the system must validate input type before attempting to parse and display a user-friendly message rather than crashing.
- 

#### **Log ID: LOG-09 (Ref: TC #009)**

- Timestamp: 2026-05-05 19:56:43
  - Module: Marks Capture / Reports
  - Description: Average calculation with multiple subjects
  - Input Data: Three subject marks entered (e.g., 75, 80, 78) (expected average: 80.00%)
  - Expected Output: Average calculated and displayed as 77.67%
  - Actual Output: System shows inconsistent calculation of averages; the result does not match expected value
  - Status: **FAIL**
  - Defect Raised: BUG#004
  - Notes: With three-mark entries of 75, 80, and 78, the expected average is 77.67%.The system failed to produce this result, confirming the average calculation logic contains an error, likely caused by integer division truncating decimal results or the report module reading from an incorrect or stale data source.
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#### **Log ID: LOG-10 (Ref: TC #010)**

- Timestamp: 2026-05-05 20:12:33
- Module: Reports
- Description: Report generation noting performance and output correctness
- Input Data: Report generated for a registered student with course and status
- Expected Output: Report displays within 3 seconds; correct student data shown
- Actual Output: System shows slow performance when loading report; report displays conflicting student name "John Doe" regardless of the actual student selected
- Status: **FAIL**
- Defect Raised: BUG#006

- **Notes:** The report loading time exceeds the acceptable 3 second performance threshold and the report displays an incorrect student name. Both defects must be resolved before the Reports module can be considered production ready.
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## Status Classification

| Metric                    | Count                   |
|---------------------------|-------------------------|
| Total Test Cases Executed | 10                      |
| Passed                    | 2                       |
| Failed                    | 8                       |
| Blocked                   | 0                       |
| Pass Rate                 | 20%                     |
| Defects Raised            | BUG#001 through BUG#006 |

## Overall Result:

The system is **NOT** ready for production deployment. Eight of ten test cases failed, identifying six distinct defects across the Login, Course Enrolment, Marks Capture, and Reports modules.

## Next Steps:

- **Defect Remediation:** The system has to undergo debugging and refactoring for all the failed test cases, and more focus has to be put on the critical bug as well as the security logic in the Login module.
- **Re-Testing:** Once debugging has been done, re-execute the test cases to verify whether the identified bugs are resolved.
- **Final QA Approval:** Compile all the corrected source code and update the execution logs to get a final deployment readiness review.